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Introduction

In modern financial markets, the accurate pricing and risk assessment of derivatives are critical for both market participants and institutional stability. As new quantitative analysts on the derivatives desk, our first mandate is to validate and verify the prices of vanilla options - European and American styles, and to compute their key sensitivities, or “Greeks.” This process forms a good foundation, before we advance to pricing exotic options, which are more complex derivatives.

The project focuses on implementing and validating option pricing models using Binomial and Trinomial tree frameworks. These discrete-time models enable the approximation of option values by simulating potential price paths of the underlying asset, incorporating market parameters such as volatility, interest rates, and time to maturity. For European options, put-call parity serves as an essential benchmark for model accuracy, providing a direct mathematical relationship between call and put prices that must hold under no-arbitrage conditions. For American options, which feature early exercise rights, the validation process extends to ensuring that their values are at least as high as those of their European counterparts.

Beyond pricing, the project includes the calculation of core Greeks such as Delta and Vega to assess the sensitivity of option prices to changes in the underlying asset price and volatility, respectively. These measures are crucial for risk management, hedging strategies, and assessing portfolio exposures. Furthermore, dynamic hedging scenarios will be explored, emphasizing the operational aspects of maintaining a neutral risk position in response to market movements.

Through this exercise, we aim to:

- Demonstrate an understanding of option pricing for both European and American contracts.
- Apply discrete-time models effectively for valuation and sensitivity analysis.
- Validate model outputs through no-arbitrage checks and cross-comparisons.
- Gain practical insights into how pricing accuracy, model assumptions, and effective hedging relate to each other.

By achieving these objectives, this work will establish a reliable framework for our desk’s option valuation processes, laying the groundwork for more advanced derivatives pricing responsibilities.

Section 1. European and American Option Pricing Using Binomial Trees

1.1 Applicability of Put-Call Parity in European Options

Put-call parity is a fundamental principle in options pricing that shows how the prices of put options and call options are linked. It says that if you hold a stock and buy a put, it should cost the same as buying a call, and investing enough to buy the stock later at the strike price.

The formula for put-call parity is

$$C - P = S_0 + Ke^{-rT}$$

Where

S_0 = current stock price

P = price of a put option

C = price of a call option

K = strike price

r = risk-free interest rate

T = time to maturity (in years)

In the context of the binomial tree framework, put–call parity applies to European options because they can only be exercised at expiration, allowing for no-arbitrage replication. The parity is derived by comparing two portfolios with identical payoffs at maturity: one consisting of a call option and a discounted bond, and another consisting of a put option and the underlying asset. Since the payoffs match, their current prices must also be equal in an arbitrage-free market (Hull, 2022).

1.2 Put–Call Parity Formula for Call Price

Starting from the standard put–call parity equation for European options:

$$C - P = S_0 + Ke^{-rT}$$

Rearranging gives:

$$C = P + S_0 - Ke^{-rT}$$

This expression shows that the price of a call is equal to the price of a put plus the spot price minus the present value of the strike. Expressing the equation this way is useful when we already know the put price (from a binomial tree or market data) and want to check the call price for consistency. In a discrete-time binomial setting, the same relationship must hold because the pricing process is based on replication (Wilmott, 2006).

1.3 Put–Call Parity Formula for Put Price

Similarly, rearranging the parity equation gives:

$$P = C - S_0 + Ke^{-rT}$$

This form is helpful when we know the call price but not the put. In practical model verification, we can compute the put price directly from the binomial tree and compare it to the value implied by this formula. If the two differ significantly beyond rounding error, it indicates a potential issue with discounting, payoff calculation, or tree construction. (McDonald, 2013).

1.4 Applicability of Put–Call Parity for American Options

Put–call parity does not generally apply to American options because they can be exercised before maturity, breaking the exact payoff equivalence that the parity relies on. The replication argument assumes identical payoffs only at expiration, but early exercise changes the value path of the options. As a result, there is no exact equality; instead, inequalities and bounds can be derived (Cox, Ross, & Rubinstein, 1979).

1.5 Pricing ATM European Call and Put Options Using a Binomial Tree

Using the parameters: S_0 = 100; r = 5%; σ = 20%; and T = 3 months, a binomial tree was built to price European call and put options. A simple code was built for pricing these options - the routine starts by setting up a discrete timeline: it splits maturity into equal steps and defines “up” and “down” multipliers for the stock, with the up move as the exponential of volatility times the square root of the step, and the down move as its reciprocal. It then derives the risk-neutral probability from the no-arbitrage condition so the discounted expected stock price grows at the risk-free rate.

Two trees are built: one for stock prices and one for option values. At maturity, the stock price tree is filled with all terminal prices, and the option’s payoff is assigned node-by-node (call: max(S–K,0), put: max(K–S,0)).

Backward induction follows: each earlier node is assigned the discounted risk-neutral expected value of its two successors. Since the option is European, only this continuation value is taken; no early-exercise check is needed.

Finally, the function outputs the option price at the root, the option-value tree, and the stock-price tree for verification and diagnostics.

For the number of time steps, different time-steps were tested in the range of 1-300, using step-size of 10. This revealed fluctuations in call option prices, which stabilizes at N=100. At that timestep, the price stabilized at $4.61 (rounded to 2 cents), which proved that N=100 was the convergence point. Same logic was applied for the put option and the price stabilized N=130 where the price was $3.37 for the remaining number of steps in the experiment. Figure 1.1 below is a convergence plot to show how prices stabilized

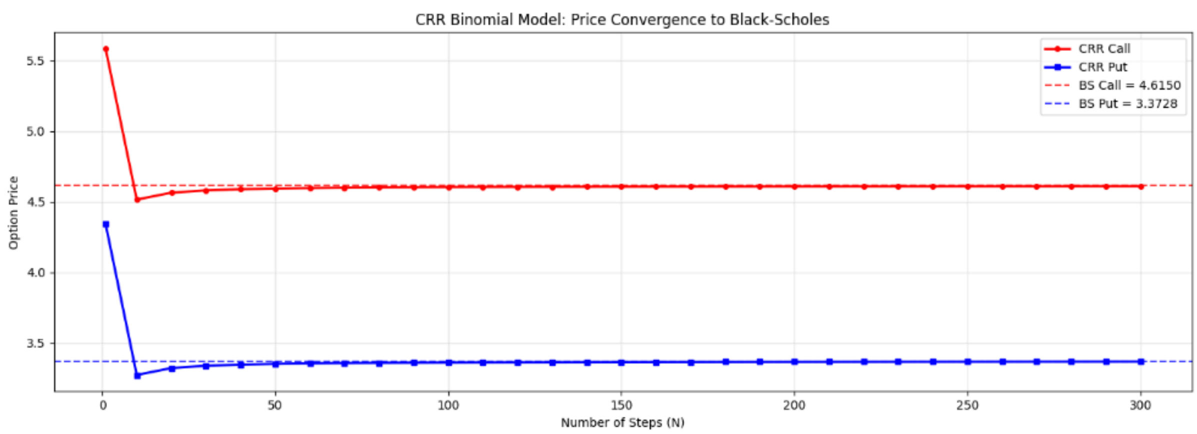

Fig 1.1 Price Convergence for European Options

1.6 Delta Computation for European Call and Put Options at Time 0

The Greek delta (Δ) was computed for both European and put options at time t=0. Delta is the sensitivity of an options price to a price change in the underlying stock

$$\Delta \approx \partial V / \partial S$$

The delta for the call is 0.57 and -0.43 for the put. The call delta is positive (the call value increases as the stock price increases) and the put delta is negative (the put value decreases as the stock price increases). Their difference is

$$\Delta c - \Delta p = 0.57 - (-0.43) = 1$$

Which means theoretical relation for European options $\Delta c - \Delta p = 1$ holds.

The European call delta is +0.57, and the put delta is −0.43. The difference in sign reflects the fact that calls and puts have opposite exposures to the underlying price:

A positive delta (the call) means the option's value rises when the underlying stock price rises. A call option benefits from upward moves in the underlying. A call is a right to buy — an increase in the stock price increases the value of that right, so Δ>0

A negative delta (the put) means the option's value falls when the underlying rises. A put option benefits from downward moves in the underlying. A put is a right to sell — an increase in S reduces the value of that right, so Δ<0

1.7 Vega Sensitivity Analysis for European Call and Put Options

Vega measures the sensitivity of an option's price to changes in the volatility of the underlying asset. Vega was computed for both the European call and put options.

For the ATM European call, the option price at 20% volatility is $4.61 and $5.59 at 25% volatility. This gives an absolute difference of 0.98. Similarly, for the European put, the option price is $3.37 at 20% and $4.35 at 25% volatility. The absolute difference of the prices is also 0.98. So, increasing volatility from 20% to 25% (a change of 0.05) increases both option prices by approximately 0.98 units. This symmetry (equal vega for put and call) often occurs in a binomial tree model especially when the options are at-the-money and have the same expiration date. This suggests that volatility effects both sides of the market similarly. Both call and put prices increase when volatility increases (vega > 0) because higher volatility raises option expected payoff.

1.8 Pricing ATM American Call and Put Options Using a Binomial Tree

Using the same parameters: S_0 = 100; r = 5%; σ = 20%; and T = 3 months, a binomial tree was built to price American call and put options.

The same code logic was also used to price American options

For the number of time-steps, different time-steps were tested in the range of 1-300, using step-size of 10. This revealed fluctuations in call option prices, which stabilizes at N=100. At that timestep, the price stabilized at $4.61 (rounded to 2 cents) up to N=300, which proved that N=100 was the convergence point. A plot of the convergence with different time steps is shown in figure 1.2 below

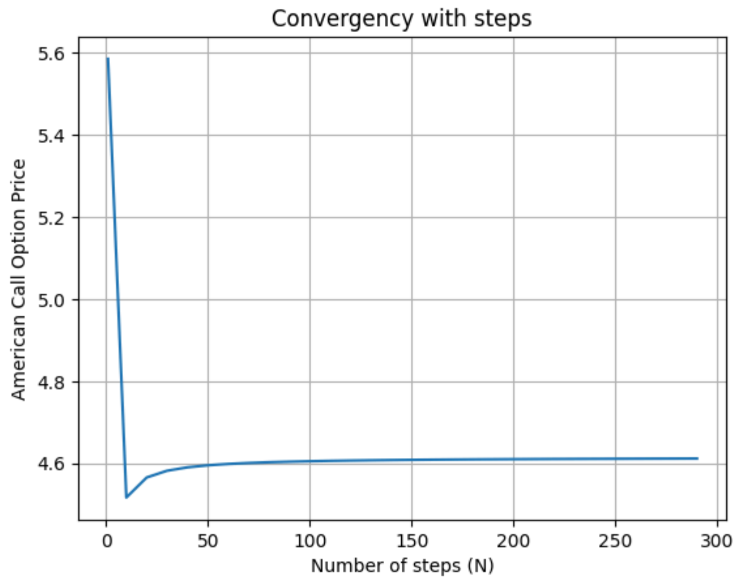

Fig 1.2 Convergence with American Call Option, Binomial Model

The same logic was applied to find the convergence point for the American put options, tested on different timesteps within the range of 1 to 300 with the step size of 10. This showed convergence at N=110, stabilizing at $3.48 as the price of the put option. Therefore, 110 was chosen as the optimal number of time steps to price the American put option. Figure 1.3 below is a visual representation of price convergence for the American put option.

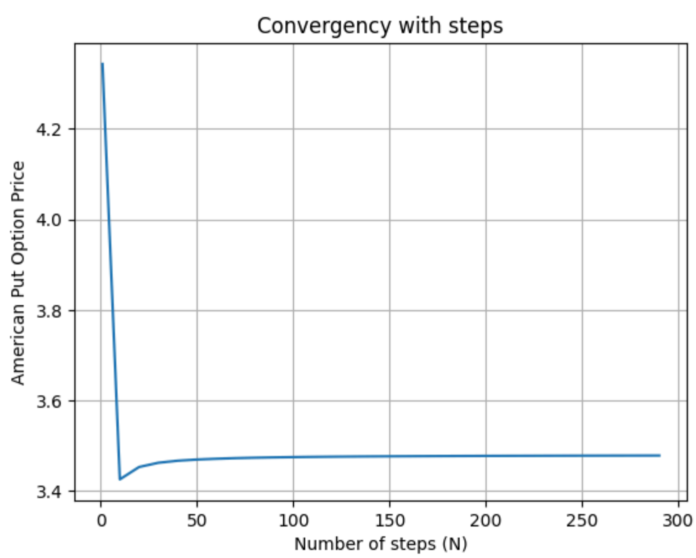

Fig 1.3 Convergence with American Put Option, Binomial Model

1.9 Delta Computation for American Call and Put Options at Time 0

The Greek delta (Δ) was computed for both American and put options at time t=0. The delta for the call is 0.57 and -0.45 for the put. The call delta is positive (the call value increases as the stock price increases) and the put delta is negative (the put value decreases as the stock price increases). Their difference is

$$\Delta c - \Delta p = 0.57 - (-0.45) = 1.02$$

which is close to the theoretical relation for European options

$$\Delta c - \Delta p = 1$$

The slight deviation is attributable to numerical approximation and the early-exercise premium for American options.

The American call delta is +0.57, and the put delta is −0.45. The difference in sign is the same behavior that was exhibited by the European options. This sign behavior holds for both American and European options, though magnitudes differ slightly due to early-exercise features and interest rate effects.

In conclusion, A delta of +0.57 for the call means that to hedge one long call, a trader would short 0.57 shares of the underlying. A delta of −0.45 for the put means that to hedge one long put, a trader would buy 0.45 shares of the underlying.

1.10 Vega Sensitivity Analysis for American Call and Put Options

Vega was computed for both the American and put options. For the ATM American call, the vega is 19.619, and for the ATM American put, the vega is 19.568. This means that if volatility increases by 1.00 (100 percentage points), the option value increases by roughly 19.6 currency units. In our case, increasing volatility from 20% to 25% (a change of 0.05) increases both option prices by approximately

$$19.6 \times 0.05 \approx 0.98$$

0.98 units. The vegas are positive for both calls and puts, as higher volatility increases the likelihood of a more favorable payoff in either direction. The near equality of the two vegas reflects that, for ATM options, volatility impacts upward and downward price movements almost symmetrically. The small difference between them arises from numerical discretization in the binomial tree and the possibility of early exercise for the American put.
The 5% increase in volatility has a very similar absolute impact on both the ATM American call and put, with both prices increasing by roughly 0.98 units. This similarity occurs because, for ATM options, the probability of large favorable movements is symmetric for calls and puts. However, the impact is slightly smaller for the put due to the possibility of early exercise, which

can cap the benefit of additional volatility — once the put is sufficiently in-the-money, early exercise might be optimal, reducing the value of waiting and thus slightly dampening the vega. For the call (assuming no dividends), early exercise is never optimal, so the option retains the full benefit of increased volatility.

1.11 Put–Call Parity Verification for European Options

Recall the formula for put-call parity is

$$C - P = S_0 + Ke^{-rT}$$

To calculate put-call parity for the European option, C = \$4.61; P = \$3.71; S_0 = \$100; K = \$100; r = 0.05; T = 0.25

Right-hand side of the formula:

$$S_0 - Ke^{-rT} = 100 - 100e^{-0.05\times0.25}$$

First, calculate the discount factor:

$$0.05 \times 0.25 = 0.0125$$

$$e^{-0.0125} \approx 0\ .98758$$

$$100 \text{ x } 0.98758 = 98.76$$

So:

$$S_0 - Ke^{-rT} \approx 100 - 98.758 = 1.242$$

Left-hand side of the formula using our prices…

$$C-P = 4.61 - 3.37 = 1.24$$

Now, let us compare both sides to check

- LHS (the prices) = 1.24
- RHS (theoretical parity) ≈ 1.242

Difference:

$$1.242 - 1.24 \approx 0.002$$

Which means the left-hand side is approximately equal to the right-hand side. So, put-call parity effectively holds for the European option as we stated in section 1.1.

Put-call parity holds because, in a frictionless and arbitrage-free market, a long call plus a short put (both European, same strike and expiry) replicates a forward contract. At expiration, one of the options exercises and the other expires worthless—resulting in receiving or paying the strike in exchange for the asset. Holding that replicated portfolio or directly holding the forward yields identical payoffs, so without arbitrage, they must have equal present values. This linkage forms the backbone of the parity relationship (Sebastien, 2010).

1.12 Put–Call Parity Verification for American Options

Checking put-call parity for our American option.

$$C - P = S_0 + Ke^{-rT}$$

Plugging in our values

- $C = 4.61$
- $P = 3.48$
- $S_0 = 100$
- $K = 100$
- $r = 5\% = 0.05$
- $T = 3$ months $= 0.25$

Right-hand side of the formula:

$$S_0 - Ke^{-rT} = 100 - 100e^{-0.05\times0.25}$$

First calculate the discount factor:

$$0.05 \times 0.25 = 0.0125$$

$$e^{-0.0125} \approx 0\ .98758$$

$$100 \times 0.98758 = 98.76$$

So:

$$S_0 - Ke^{-rT} \approx 100 - 98.758 = 1.242$$

Left hand side of the formula using our prices…

$$C - P = 4.61 - 3.48 = 1.13$$

Now let us compare both sides to check

- LHS (the prices) = 1.13
- RHS (theoretical parity) ≈ 1.242

Difference:

$$1.242 - 1.13 \approx 0.112$$

The prices do not perfectly satisfy put–call parity, with a deviation of about 0.11. This is a practical proof of what we stated in section 1.4.

Put-call parity does not strictly hold for American options because early exercise is possible. The arbitrage arguments behind parity assume both options are held until maturity (as in the European case) (Whaley, 1981).

1.13 Price Comparison: European vs. American Call Options

Both European and American call options are the same; priced at $4.61. For non-dividend-paying stocks, American calls never exceed their European counterparts; the equality holds because early exercise sacrifices time value without any compensating benefit.

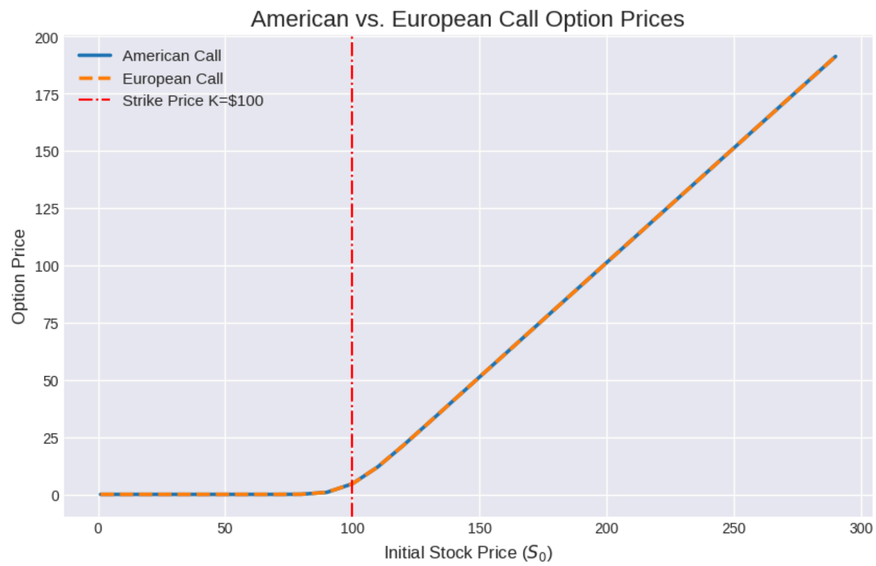

Fig 1.4 American vs European Call Option Prices

1.14 Price Comparison: European vs. American Put Options

The American put is worth more ($3.48) than the European put ($3.37), which matches theory: unlike calls, early exercise of a put can be optimal (Hull, 2021).

Fig 1.5 American vs European Put Option Prices

Section 2: Option Pricing Across Moneyness Level Using Trinomial Trees

2.1 European Call Option Pricing Across Five Strike Levels Using a Trinomial Tree

Using the same parameters as in section 1, the price of European options were calculated using the Trinomial model. The model calculated option prices for options which are Deep OTM, OTM, ATM, ITM, and Deep ITM. To accomplish this, different strike prices with moneyness of 90%, 95%, ATM, 105%, 110% were used.

A Python function was used which builds a recombining trinomial tree for the underlying asset, using volatility to determine up and down moves and risk-neutral probabilities for up, down, and middle moves. It then calculates option payoffs at maturity for all given strike prices and works backwards step-by-step, discounting the expected values under the risk-neutral probabilities. For European options, the final value at the root of the tree is the option price for each strike.

For the European call option, Table 2.1 below shows the option prices across different moneyness levels.

Table 2.1 European Call Option Prices at Different Strike Prices

Strike($)	Moneyness (%)	Call Price ($)	Classification
90	11.11	11.67	Deep ITM
95	5.26	7.72	ITM
100	0.00	4.61	ATM
105	-4.76	2.48	OTM
110	-9.09	1.19	Deep OTM

From the table, we can see that as the strike price increases, the price of the call decreases. It makes sense because a higher strike price means the buyer has to pay more to exercise the call, so the likelihood and magnitude of a profitable payoff drops. As the strike moves further above the current stock price, the option becomes less likely to finish in-the-money, reducing its expected value. This inverse relationship between strike and call price is a direct result of lower intrinsic value and reduced probability of exercising profitably.

2.2 European Put Option Pricing Across Five Strike Levels Using a Trinomial Tree

The same was also applied for assessing moneyness for puts at the same strike prices of $90, $95, $100, $105 and $110, and the result is summarized in table 2.2 below.

Table 2.2 European Put Option Prices at Different Strike Prices

Strike($)	**Moneyness (%)**	**Put Price ($)**	**Classification**
90	-10.00	0.55	Deep OTM
95	-5.00	1.53	OTM
100	0.00	3.36	ATM
105	5.00	6.17	ITM
110	10.00	9.83	Deep ITM

We can see that the price of the option, put in this case increases as the strike prices increase, which is unlike the inverse relationship as was the case of the the European call option. This makes sense because a higher strike price increases the amount the holder could receive if the option finishes in-the-money, raising both the potential payoff and the probability of exercising profitably, which increases the put's value.

2.3 American Call Option Pricing Across Five Strike Levels Using a Trinomial Tree

American call option was price across the 5 different strike prices used to price the European options ($90, $95, $100, $105 and $110), and the result of the prices with different strike prices and their moneyness is represented in Table 2.3 below

Table 2.3 American Call Option Prices at Different Strike Prices

Strike($)	**Moneyness**	**Call Price ($)**	**Classification**

	(%)		
90	11.11	11.67	Deep ITM
95	5.26	7.72	ITM
100	0.00	4.61	ATM
105	-4.76	2.48	OTM
110	-9.09	1.19	Deep OTM

Just like it's European counterpart, as the strike price of the American call increases, the price of the call decreases, and the same explanation of higher strike price meaning the buyer has to pay more to exercise the call applies.

2.4 American Put Option Pricing Across Five Strike Levels Using a Trinomial Tree

American put options was also priced across the same 5 moneyness levels and the result is displayed in Table 2.4 below

Table 2.4 American Put Option Prices at Different Strike Prices

Strike	**Moneyness (%)**	**Put Price**	**Classification**
90	-10.0	0.56	Deep OTM
95	-5.0	1.57	OTM
100	0.0	3.47	ATM
105	5.0	6.42	ITM
110	10.0	10.33	Deep ITM

2.5 Options Graphs

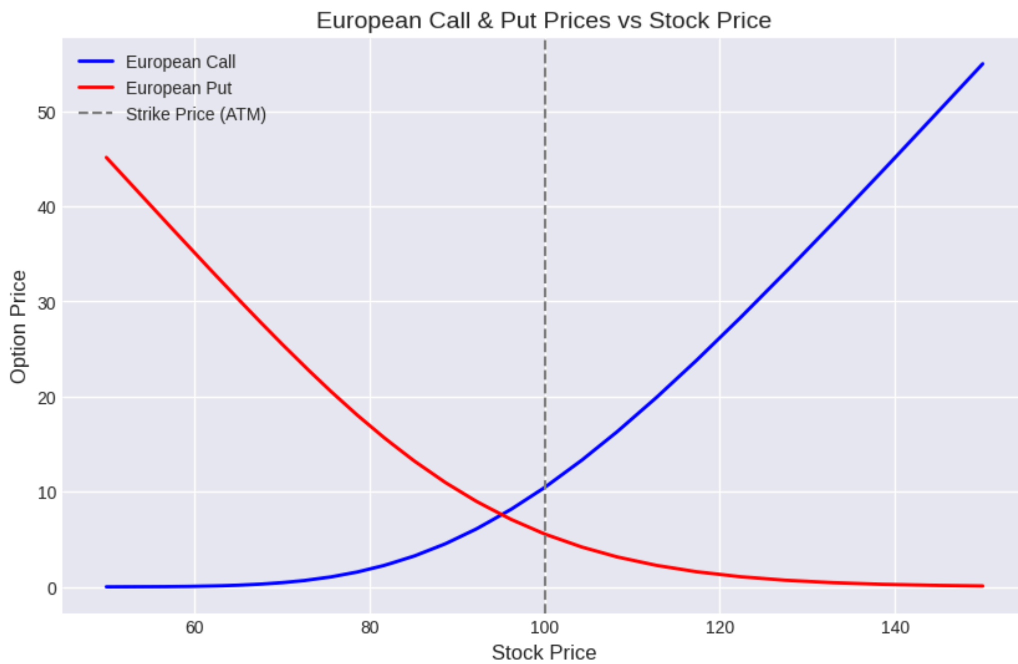

Fig 2.1 European Call & Put Price Trends vs Stock Price

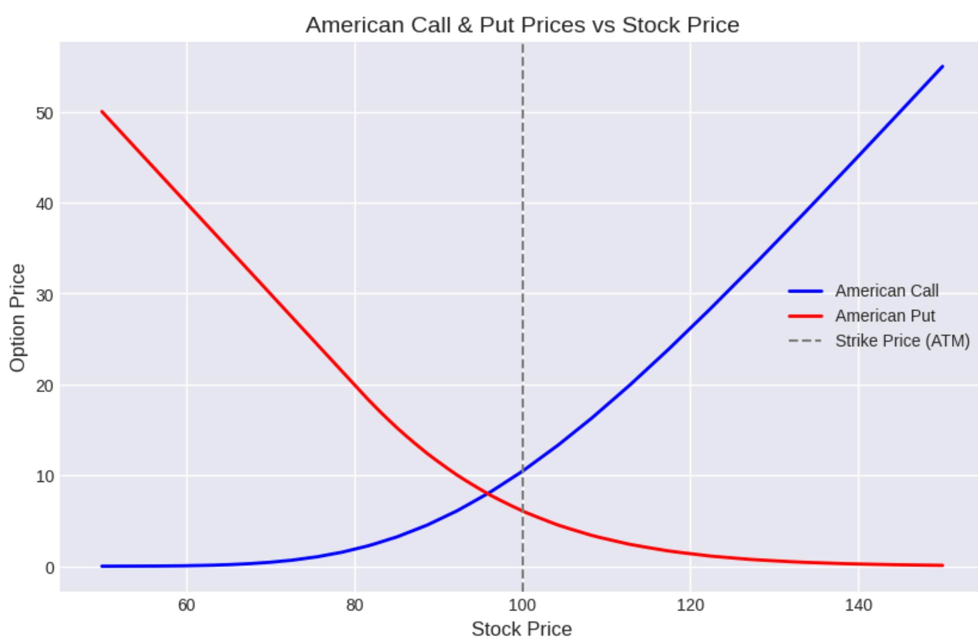

Fig. 2.2 American Call and Put Price Trends Versus Stock Price

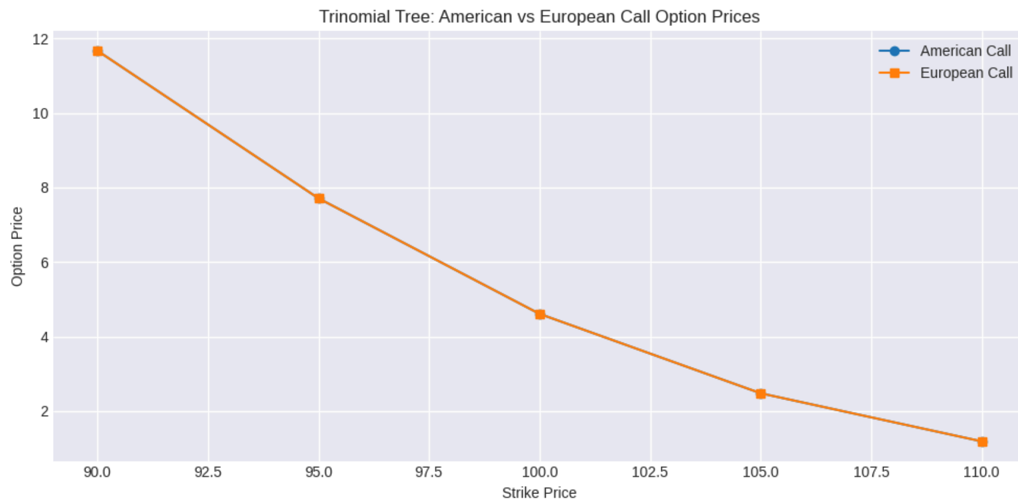

Fig 2.3 Comparative Price Trends for European and American Calls Versus Strike Price

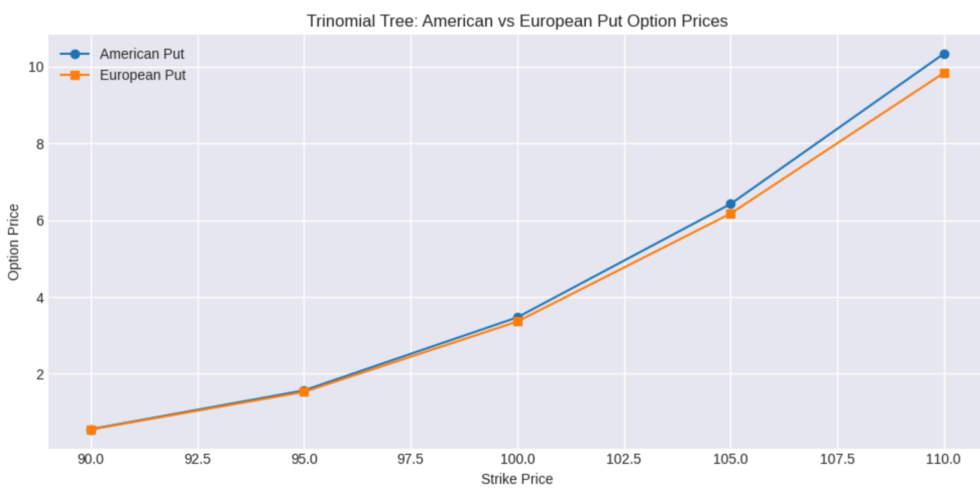

Fig 2.4 Comparative Price Trends for European and American Puts Versus Strike Price

2.9 Put-Call Parity Verification for European Options Across Multiple Strikes

Python code was used to calculate put-call parity for European option across the different levels of moneyness. Table 2.5 below is the result of the calculation.

Table 2.5 Put-Call Parity for European Options Across Multiple Strikes

Strike (K)	**Call C**	**Put P**	**C − P**	**$S_0 - Ke^{-rT}$**	**Difference (LHS−RHS)**	**Parity Holds?**
90	11.67	0.55	11.12	11.12	0.00	Yes
95	7.72	1.53	6.19	6.18	0.01	Yes
100	4.61	3.36	1.25	1.24	0.01	Yes
105	2.48	6.17	−3.69	−3.70	0.01	Yes
110	1.19	9.83	−8.64	−8.63	−0.01	Yes

At any strike K, a long call + discounted bond (Ke^{-rT}) replicates the exact payoff of a long put + stock at maturity. Since these two portfolios have the same payoff in every state of the world, their prices today must also be equal otherwise, arbitrage would exist.

2.10 Put-Call Parity Verification for American Options Across Multiple Strikes

We also checked put-call aprity for American options and the results are represented in Table 2.6 below

Table 2.6 Put-Call Parity for American Options Across Multiple Strikes

Strike (K)	**Call (C)**	**Put (P)**	**C – P**	**$S_0 - K \cdot \exp(-rT)$**	**Difference**	**Parity Holds?**
90	11.67	0.56	11.11	11.1180	–0.0080	True
95	7.72	1.57	6.15	6.1801	–0.0301	False
100	4.61	3.47	1.14	1.2422	–0.1022	False
105	2.48	6.42	–3.94	–3.6957	–0.2443	False

110	1.19	10.33	–9.14	–8.6336	–0.5064	False

Put–call parity is derived under the assumption that options can only be exercised at maturity (European style). American options, however, give the holder the right to exercise early. This extra flexibility affects pricing, particularly for puts: an American put can be exercised early when deep in-the-money, making it more valuable than its European counterpart. Therefore, the exact equality in put–call parity breaks down for American options. Instead, only inequalities hold.
At strike 90, both the call and the put are out-of-the-money (especially the put). Early exercise provides no additional value since exercising before maturity would still result in no payoff. In such cases, the American and European option values converge, so put–call parity approximately holds.

Section 3: Dynamic Delta Hedging and Comparative Option Analysis

3.1 European Put Option Pricing Using a 3-Step Binomial Tree

We will now calculate the price of a European put option with the given parameters below

Given parameters:

- S_0 = $180
- r = 2% = 0.02
- σ = 25% = 0.25
- T = 6 months = 0.5 years
- K = $182
- n = 3 steps

The first step is to calculate the binomial parameters: the change in time (Δt), upward factor (u), downward factor (d), and risk-neutral probabilities (p)

$\Delta t = T/n = 0.5/3 = 0.1667$

$u = e^{\sigma\sqrt{\Delta t}} = e^{0.25 \times \sqrt{0.1667}} = e^{0.25 \times 0.408} = e^{0.102} = 1.107$

$d = 1/u = 1/1.107 = 0.903$

$p = e^{r\Delta t} - d / u - d = e^{0.02 \times 0.1667} - 0.903 / 1.107 - 0.903 = 1.003 - 0.903 / 0.204 = 0.1/0.204 = 0.49$

This means that the risk-neutral probability of an upward move is 0.49, meaning the risk-neutral probability of a downward move is 1-0.49 = 0.51

Next, we compute each price path

Step 0 (Initial Price)

$S_0 = 180$

Step 1:

{u} =180×1.107 = 199.26

{d} = 180 × 0.903 = 162.54

Step 2:

{u,u} = 199.26 × 1.107 = 220.58

{u,d} = 199.26 × 0.903 = 179.93

{d,d} = 162.64 × 0.9 = 146.86

Step 3:

{u,u,u} = 220.58 × 1.107 = 244.18

{u,u,d} = 220.58 × 0.903 = 199.18

{u,d,d} = 179.93 × 0.903 = 162.48

{d,d,d} = 146.86 × 0.903 = 132.61

The price paths are visualized in figure 3.1 below

Fig 3.1 Price path for European Option at $S_0 = 180$

The next step is to calculate the put option values at expiration (payoff).

Since our strike price K = $182

Put payoff = max (K - S, 0) = max (182 - S, 0)

At 244.18: Payoff = max (182 - 244.18, 0) = 0

At 199.18: Payoff = max (182 - 199.18, 0) = 0

At 162.48: Payoff = max (182 - 162.48, 0) = 19.52

At 131.61: Payoff = max (182 - 131.61, 0) = 50.39

Next, we work backwards through the tree

$V = e^{-r\Delta t} \times [pV_{up} + (1-p)\, V_{down})$

With

$e^{-r\Delta t} = e^{-0.02 \times 0.1667} = e^{-0.0033} = 0.9967$

and p = 0.49

At Step 2:

- **At {u,u}, S = 220.58**: 0.9967(0.49 × 0 + 0.51 × 0) = 0.00
- **At {u,d}, S = 179.93**: 0.9967(0.49 × 0 + 0.51 × 19.52) = 0.9967 × 9.9552 = 9.92
- **At {d,d}, S = 146.86**: 0.9967(0.49 × 19.52 + 0.51 × 50.39) = 0.9967 × 35.263 = 35.15

At Step 1:

- **At {u}, S = 199.26**: children are {u,u} and {u,d} with values 0.00 and 9.92

  V_u = 0.9967(0.49 × 0.00 + 0.51 × 9.92) = 0.9967 × 5.0592 = 5.04

- **At {d}, S = 162.54:** children are {u,d} and {d,d} with values 9.92 and 35.15

  V_d = 0.99667(0.49 × 9.92 + 0.51 × 35.15) = 0.9967 × 22.7873 = 22.71

At Step 0: Root price

Children are {u} and {d} with values 5.04 and 22.71:

$Price_0$ = 0.9967(0.49 × 5.04 + 0.51 × 22.71) = 0.9967 × 14.0517 = 14.005

The European put option is $14.01

3.2 Delta Hedging Process for a Selected Path in the European Put Option Tree

For the hedging, we chose the path {d,u,u}

First step would be to compute delta along the path

Formula for delta at a node

$\Delta = V_{up} - V_{down} / S_{up} - S_{down}$

We'll compute the deltas at nodes encountered on the {d,u,u} path:

At time 0 (root)

- $S_U = 199.26$, $V_U = 5.04$
- $S_d = 162.54$, $V_d = 22.71$

$\Delta_0 = 5.04 - 22.71 / 199.26 - 162.54 = -17.67 / 36.72 = -0.48$

At time 1 (after {d}, node {d} at S = 162.00)

Children: {u,d} and {d,d} with values V_{ud}= 9.92 and V_{dd}=35.15 and stock prices S_{ud}=179.93, S_{dd}=146.86

$\Delta_d = 9.92 - 35.15 / 179.93 - 146.86 = -25.23 / 32.40 = -0.78$

At time 2 (after {d,u}, node {u,d} at S = 178.20)

Children: {u,d,u} and {u,d,d} with values V_{udu}= 0.0 and V_{udd}=19.52, stock prices S_{udu}=199.18, S_{udd}=162.48

$\Delta_{ud} = 0.00 - 19.52 / 199.18 - 162.48 = -19.52 / 36.7 = -0.52$

$\Delta_0 = -0.48$, $\Delta_d = -0.78$, $\Delta_{ud} = -0.52$

Hedging rule for shorting the put

- At t=0 (after selling the put) we establish the initial hedge by shorting 0.48 shares. The proceeds from shorting $0.48 \times S_0$ are added to the cash account (together with the premium received).
- When the tree moves to node {d,d} and the option delta becomes −0.78, we increase the short position from −0.48 to −0.78: short an additional 0.30 shares and add the proceeds to cash.
- When the tree then moves to node {d,u} and the delta reduces in magnitude to −0.52, we decrease the short position from −0.78 to −0.52: buy back 0.26 shares and pay cash for them.

- At maturity {d,u,u}, the stock is S=199.18, which is above the strike 182. The put expires worthless. At this point, we close the hedge by buying back the remaining 0.52 shares we are short. No option payoff is due, so the cash account balance (which has been compounding at the risk-free rate) remains as profit.

Table 3.1 Cash Account Evolution for European Put Delta Hedge

Time	**t = 0**	**t = 1 {d}**	**t = 2 {d,u}**	**t = 3 ({d,u,u}, maturity)**
Underlying price	\$180.00	\$162.54	\$179.93	\$199.18
Put option value (long)	\$14.01	\$22.71	\$9.92	\$0.00
Δ (long put)	-0.48	-0.78	-0.52	—
Shares held (seller)	-0.48	-0.78	-0.52	0.00
Stock portfolio value (shares × price)	-\$86.40	-\$126.54	-\$93.56	\$0.00
Cash before trade (start of period, grown at r)	\$0.00	\$100.75	\$150.01	\$103.58
Trade at this node (buy + / sell -)	**+\$100.41** (premium 14.01 + short 0.48×180=86.40)	**+\$48.76** (short 0.30 at 162.54)	**-\$46.78** (buy 0.26 at 179.93)	**-\$103.57** (buy 0.52 at 199.18)
Cash after trade	\$100.41	\$149.51	\$103.23	\$0.01

Net portfolio value (Cash after trade + stock portfolio value + liability from long put)	$0.00	$0.02	$-0.25	$0.00

3.4 American Put Option Pricing Using a 25-Step Binomial Tree

Using the same parameters for pricing the European put and calculating the hedge at every node, an American put option was priced using a binomial tree model. The price of the option premium was calculated to be $13.04 and the delta at each node was:

Step 0: -0.48

Step 1: -0.40; -0.56

Step 2: -0.32; -0.48; -0.65

Step 3: -0.24; -0.40; -0.57; -0.73

Step 4: -0.18; -0.31; -0.48; -0.66; -0.81

We decided to choose the path {d,d,d,d…} until an early exercise (or not) was appropriate. Below is the cash account evolution throughout the different steps of this path.

Table 3.2 Cash Account Evolution for a Selected Path in the American Put Option Tree

Time step	**Node (i,j)**	**Underlying price**	**Option value (long)**	**Δ (long put)**	**Shares held (seller)**	**Stock portfolio value**	**Cash before trade**	**Trade cash (inflow + / outflow -)**	**Cash after trade**
0	(0,0)	180.00	13.04	-0.4756	-0.4756	-85.60	0.00	98.64	98.64
1	(1,1)	173.75	16.05	-0.5608	-0.5608	-97.44	98.68	14.81	113.49

2	(2,2)	167.71	19.48	-0.6479	-0.6479	-108.66	113.53	14.60	128.14
3	(3,3)	161.89	23.30	-0.7331	-0.7331	-118.67	128.19	13.79	141.98
4	(4,4)	156.26	27.48	-0.8124	-0.8124	-126.94	142.03	12.39	154.43
5	(5,5)	150.83	31.96	-0.8822	-0.8822	-133.07	154.49	10.54	165.03
6	(6,6)	145.59	36.64	-0.9400	-0.9400	-136.86	165.09	8.41	173.51
7	(7,7)	140.54	41.47	-0.9844	-0.9844	-138.34	173.58	6.23	179.81
8	(8,8)	135.65	46.35	—	0.0000	0.00	179.88	-133.53	0.00

Early exercise occurred at step 8 with payoff 46.35.

3.5 Comparative Analysis of Delta Hedging for European and American Put Options

Behavior of the Delta (Δ)

- European Put: The Delta evolves smoothly based on how the cost is supposed to replicate in future. On a downward path, the Delta becomes more negative relatively predictably as the option goes deeper ITM.
- American Put: The Delta's behavior is constrained because of the early exercise feature. As the stock price falls, the option quickly loses time value.

Frequency and Magnitude of Hedge Rebalancing

- European Put: Rebalancing is mostly driven by the changes in the theoretical delta due to how the price moved and also time decay. The adjustments are continuous and calculable until expiration.
- American Put: As the hedgers, we must be prepared for a discrete, large-scale rebalancing event at any time. If the stock price drops to a level where we should trigger an early exercise, our short option position is suddenly assigned.

Hedging a short American option position is inherently riskier and more complex than hedging its European counterpart. The critical risk is not just the direction of the underlying move but also the timing of the resulting payoff.

3.6 Pricing and Delta Hedging for an Asian ATM Put Option

We used Python to price an Asian ATM put option using Monte Carlo simulation and then simulate the delta hedging process along a specific path (all down moves). We also constructed a cash account evolution table in the same format.

Table 3.3 Cash Account Evolution for Asian Put Option

Time	Underlying Price	Put Option Value	Delta	Shares Held	Stock Portfolio Value	Cash Before Trade	Trade at This Node	Cash After Trade	Net Portfolio Value
t = 0	$180.00	$0.0000	-0.44	-0.44	-79.69	$0.0000	$86.48	$86.48	$6.79
t = 1	$154.60	$12.58	-0.44	-0.44	-68.45	$86.51	-0.0000	$86.51	$5.49
t = 2	$153.57	$17.12	-0.44	-0.44	-67.99	$86.55	-0.0000	$86.55	$1.44
t = 3	$153.11	$19.51	-0.44	-0.44	-67.78	$86.58	-0.0000	$86.58	-0.71
t = 4	$152.35	$21.09	-0.44	-0.44	-67.45	$86.62	-0.0000	$86.62	-1.93

t = 5	$158.34	$21.17	-0.44	-0.44	-70.10	$86.65	-0.0000	$86.65	-4.62
t = 6	$160.34	$20.94	-0.44	-0.44	-70.99	$86.68	-0.0000	$86.68	-5.24
t = 7	$148.22	$22.27	-0.44	-0.44	-65.62	$86.72	-0.0000	$86.72	-1.17
t = 8	$183.58	$19.41	-0.44	-0.44	-81.27	$86.75	-0.0000	$86.75	-13.93
t = 9	$168.72	$18.60	-0.44	-0.44	-74.69	$86.79	-0.0000	$86.79	-6.50
t = 10	$159.52	$18.76	-0.44	-0.44	-70.62	$86.82	-0.0000	$86.82	-2.56
t = 11	$135.12	$20.93	-0.44	-0.44	-59.82	$86.86	-0.0000	$86.86	$6.11
t = 12	$137.29	$22.59	-0.44	-0.44	-60.78	$86.89	-0.0000	$86.89	$3.52
t = 13	$200.71	$19.52	-0.44	-0.44	-88.86	$86.93	-0.0000	$86.93	-21.44
t = 14	$172.66	$18.71	-0.44	-0.44	-76.44	$86.96	-0.0000	$86.96	-8.19
t = 15	$136.42	$20.26	-0.44	-0.44	-60.40	$87.00	-0.0000	$87.00	$6.34

t = 16	$137.62	$21.56	-0.44	-0.44	-60.93	$87.03	-0.0000	$87.03	$4.55
t = 17	$191.81	$19.72	-0.44	-0.44	-84.92	$87.07	-0.0000	$87.07	-17.57
t = 18	$180.47	$18.66	-0.44	-0.44	-79.89	$87.10	-0.0000	$87.10	-11.45
t = 19	$136.33	$19.91	-0.44	-0.44	-60.36	$87.14	-0.0000	$87.14	$6.87
t = 20	$170.27	$19.43	-0.44	-0.44	-75.38	$87.17	-0.0000	$87.17	-7.65
t = 21	$140.90	$20.33	-0.44	-0.44	-62.38	$87.21	-0.0000	$87.21	$4.50
t = 22	$175.99	$19.63	-0.44	-0.44	-77.91	$87.24	-0.0000	$87.24	-10.30
t = 23	$132.81	$20.79	-0.44	-0.44	-58.80	$87.28	-0.0000	$87.28	$7.69
t = 24	$207.86	$18.85	-0.44	-0.44	-92.02	$87.31	-0.0000	$87.31	-23.56
t = 25	$136.71	$19.80	-0.44	-0.44	-60.52	$87.35	-0.0000	$87.35	$7.03

3.7 Comparative Analysis of Asian ATM Put and American Put Options

We observed that the Asian Put had stable delta values (-0.44 consistently) throughout the entire path, while the American Put was not really stable. The delta evolved, increasing from

-0.48 to -0.98 as the underlying declined. This implies that the Asian option needs less frequent rebalancing, reducing transaction costs.

The cash account of the Asian option grows steadily from $86.48 to $87.35, showing low volatility, while the American put's cash account shows more dynamic evolution, growing from $98.64 to $179.88 before exercise. This implies that Asian option hedging requires less capital allocation to the cash account.

The net portfolio value of the Asian put fluctuates significantly (-$23.56 to +$7.69), indicating that the hedging was imperfect, while the American put's hedging was near-perfect with net portfolio values close to zero until exercise. This implies that American option delta hedging provides more effective risk mitigation despite higher rebalancing needs

The Asian put option demonstrates better operational efficiency in delta hedging due to the stability of its delta, resulting in lower transaction costs and simpler implementation.

The American put option, while requiring more active management and higher transaction costs, provides substantially better risk mitigation through more responsive delta adjustments. The early exercise feature provides a clear hedging termination point, though it introduces additional exercise timing risk.

For risk managers, the choice between these instruments should consider:

1. The trade-off between operational efficiency and hedging precision

2. The expected market volatility and direction

3. The capacity for active position management

4. Transaction cost constraints

The results suggest that while Asian options offer simplified hedging strategies, their path-dependent nature creates challenges in achieving precise delta neutrality. American options, despite their complexity (due to early exercise), provide more reliable hedging performance for risk-averse portfolios.

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